MORTIS: EFFICIENT COMPILER-GUIDED LLM AGENT SERVING VIA INTRA-PROMPT HETEROGENEITY

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ABSTRACT

Existing agent serving systems schedule the KV-cache at a coarse binary prefix–suffix granularity. We observe that agent prompts exhibit finer schedulable structure, which we term *intra-prompt heterogeneity*: they can be decomposed into multiple heterogeneous components that differ in sharing pattern and lifetime. This property creates opportunities to enhance KV-cache management. Meanwhile, frameworks that let LLM invocations be declared as language constructs emit prompts programmatically, making intra-prompt heterogeneity analyzable from the program itself. To this end, we present MORTIS, a compiler-serving co-design that recovers and exploits intra-prompt heterogeneity for efficient KV-cache management. Our evaluation shows that MORTIS achieves up to 1.99× JCT speedup and 1.97× throughput speedup over SGLang, with geometric means of 1.45× and 1.49×.

1 INTRODUCTION

Agent applications accomplish tasks through a series of dependent LLM requests that often share long contexts. A recent characterization reports that 85–97% of prompt tokens are shared across requests (Srivatsa et al., 2025). Serving one LLM request involves a prefill phase that computes the key-value (KV) cache for the prompt, followed by autoregressive decoding. Prefill is particularly expensive for long prompts because attention cost grows quadratically with prompt length, but the extensive context sharing in agent applications creates substantial opportunities to avoid redundant prefill computation by reusing the KV-cache. However, KV-cache reuse cannot be taken for granted: limited GPU memory means that not every reusable KV-cache can be retained. Agent serving systems must therefore manage the KV-cache carefully, as losing reusable entries is costly: once the KV-cache thrashes, per-request prefill time can grow by 6.8× and generation throughput can drop by up to 66% (Yuan et al., 2026).

To efficiently manage the KV-cache, agent serving systems such as KVFlow, Helium, PBKV, and CacheScout (Pan et al., 2025; Wadlom et al., 2026; Zheng et al., 2026; Zhang et al., 2026) exploit agent workload predictability to inform scheduling decisions. However, these systems treat prompts through a coarse binary view: a reusable prefix followed by an opaque main prompt (suffix). These systems' scheduling operations are conducted on the identified prefix, making them fail to fully exploit the recurring behavior patterns and extensive context sharing of agent workloads, as they barely consider information inside prompts.

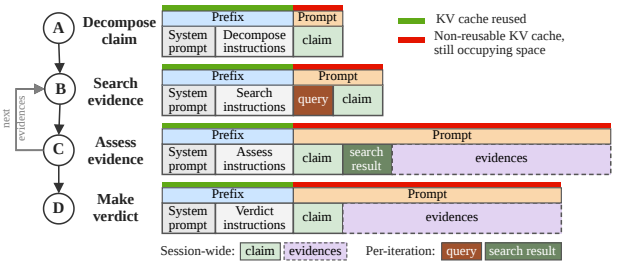


Figure 1. A Fact-check agent that verifies a *claim*'s correctness, whose prompts can be decomposed into heterogeneous components with distinct sharing patterns and lifetimes, exposing fine-grained scheduling opportunities.

We observe that agent prompts have a finer structure that can guide scheduling: prompts can be decomposed into heterogeneous components whose **sharing pattern** and **lifetime** differ. We call this property **intra-prompt heterogeneity**. The Fact-check agent in Figure 1 exhibits it: **(i)** *sharing pattern*: *claim* is shared by all four calls, whereas *evidences* is shared only by calls C and D; **(ii)** *lifetime*: *claim* and *evidences* live for the whole session, with *evidences* growing every iteration, whereas *query* and *search_result* are replaced in each loop iteration.

This property opens two optimization opportunities. **First**, more prefix cache reuse can be exposed by laying out components within prompts. Because prefix reuse stops at the first different token, placing stable, widely shared compo-

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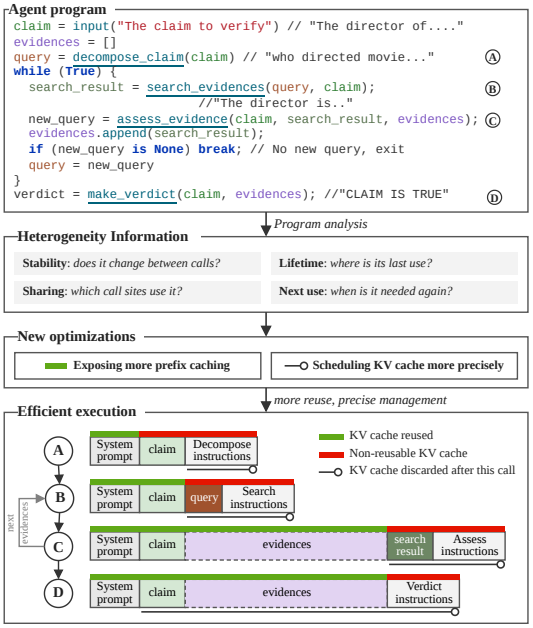


Figure 2. Intra-prompt heterogeneity is analyzable from code, and this property can be exploited for more efficient serving.

nents at the front keeps a divergent token from cutting off the reusable prefix, such as placing shared *claim* at the beginning. **Second**, the KV-cache can be scheduled more precisely, at the granularity of components rather than as one monolithic prompt. For example, the KV-caches of *query* and *search_result* are dead after their last use and should be evicted.

Exploiting these opportunities requires knowledge of intra-prompt heterogeneity. **This knowledge resides in the agent program itself and can be recovered via compiler techniques.** Agent frameworks such as byLLM, NOOA, and DSPy (Furgale et al., 2026; Khattab et al., 2024; Dantanarayana et al., 2025) expose LLM invocations through programming abstractions, letting developers declare them as functions with typed parameters. The framework runtime then generates the prompts from these declarations and the argument values. Consequently, components within prompts are mapped to variables, so sharing patterns and lifetimes are encoded in the program. Intra-prompt heterogeneity is therefore analyzable (Figure 2).

To this end, we present MORTIS, a compiler–serving co-design that recovers and exploits intra-prompt heterogeneity for efficient agent serving. MORTIS introduces *adaptive prompt re-layout* and *proactive KV-cache scheduling* to maximize prefix-cache reuse and to schedule scarce cache memory more efficiently. However, recovering and exploiting this heterogeneity poses several challenges. **First, comprehensive recovery requires both code information and actual runtime behaviors.** Static analysis can provide the basic structure, but knowledge of behavior comes from the runtime. MORTIS bridges the gap by combining information from both sides with its compiler-serving co-design. **Second, prompt layout is local to each call-site, but maximizing prefix reuse requires global coordination.** MORTIS uses a five-level tagging system and workflow graph analysis to determine the final layout that maximizes cache reuse. **Third, KV-cache scheduling consumes the resources it tries to save.** Retention and prefetching hold memory that other requests could use, yet evicting too eagerly forces recomputation. MORTIS resolves this tension by budgeting retained prefixes and prefetching only into memory reclaimed from dead cache entries.

In summary, we make the following contributions: **(i)** We identify intra-prompt heterogeneity in agentic workloads and the optimization opportunities it creates. **(ii)** We propose a compiler analysis methodology that recovers intra-prompt heterogeneity to guide efficient KV-cache management. **(iii)** We design and implement MORTIS, a compiler-serving co-design that exploits recovered intra-prompt heterogeneity for efficient KV-cache scheduling. Our evaluation shows that MORTIS achieves geometric-mean speedups of 1.45× in average JCT and 1.49× in throughput over SGLang, and up to 1.99× and 1.97× respectively. On a mixed workload, the geometric-mean speedups are 1.53× in JCT and 1.56× in throughput, and up to 1.83× and 1.86×.

2 BACKGROUND AND MOTIVATION

Prefix caching and KV-cache tiering Serving engines such as SGLang (Zheng et al., 2024) and vLLM (Kwon et al., 2023) reuse the longest matching prefix, computing only the remaining tokens. Tiered caching systems such as CachedAttention (Gao et al., 2024), Mooncake (Qin et al., 2025), and SGLang HiCache (SGLang Contributors, 2026) extend the KV-cache beyond device memory with host memory and optionally disk storage to improve capacity. However, KV-cache that is not in the device tier must be promoted back to device before use.

KV-cache management for LLM agents Agent serving systems use recurring patterns in agentic workloads to manage the KV-cache. KVFlow (Pan et al., 2025) derives each agent's steps-to-execution from the workflow to guide eviction and prefetching of each agent's prompt prefix; Helium (Wadlom et al., 2026) analyzes the workflow to order LLM calls and prewarm static prefixes; Learned Prefix Caching (Yang et al., 2025), PBKV (Zheng et al., 2026) and CacheScout (Zhang et al., 2026) predict upcoming agent invocations with trained models to decide which caches to retain. Continuum (Li et al., 2025) pins the KV-cache across tool gaps; KVTether (Fu et al., 2026) preferentially preserves live-but-idle KV-cache that is waiting for reuse in

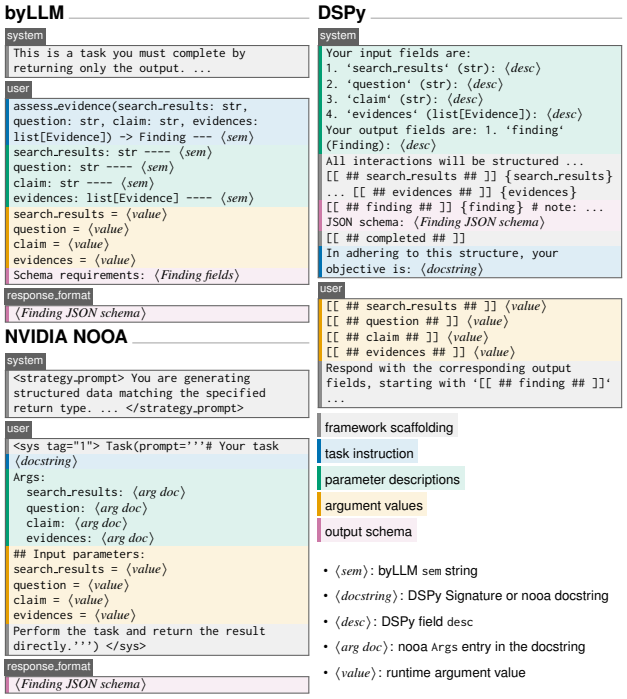


Figure 3. Prompts lowered by three frameworks for the same call share five kinds of components; only their order differs.

ReAct agents. They conduct management operations like prefetching, eviction, and protection on the identified prefix cache. Some systems, such as Pythia (Yu et al., 2026), furthermore prefill the predicted next invocations to proactively prepare KV-cache that does not exist yet.

Frameworks with LLM calls as typed functions A class of agent frameworks exposes model-backed computation through typed programming interfaces. Examples include byLLM's meaning-typed programming (MTP) abstraction (Dantanarayana et al., 2025), DSPy's Signatures and Modules (Khattab et al., 2024), and NOOA's typed Python methods (Furgale et al., 2026). The developer declares a function or signature describing the desired behavior, inputs, and return type, and invokes it like an ordinary call within the program's control flow. At each invocation, the framework runtime lowers the declaration and its argument values into messages that are sent to the model.

Although these agent frameworks differ in surface syntax, they lower LLM invocations in similar ways. Developers declare LLM invocations as functions or function-like declarations, and the framework serializes that declaration together with the runtime arguments into a single prompt. Their generated prompts carry the same functional components. As shown in Figure 3, these frameworks emit the same five kinds of components: framework scaffolding, task instruction, parameter descriptions, argument values, and output schema. The only difference is their relative positions.

Scheduling Opportunity: The order of the five components is a framework convention: the runtime applies the same template to every call. Moreover, the order of argument values is just an artifact of the developer's personal preference: it follows the order of the argument list in the code. Therefore, this layout can be managed for better performance. For example, DSPy's static adapter (DSPy Contributors, 2026) lets users tweak the lowering process to put stable prompt components at the front. NOOA (Furgale et al., 2026) lets developers label arguments as static prefix, append-only history or dynamic suffix so that the framework can order them for better cache reuse.

Intra-prompt heterogeneity in programs Prompts can be decomposed into components with different sharing patterns and lifetimes, and frameworks with LLM calls as typed functions offer structural guarantees to make this information analyzable. Because components are mapped to program variables, the semantic boundaries and relationships are therefore analyzable from the program.

Scheduling Opportunity: With the analyzable sharing patterns and lifetimes, we can determine the operations for specific KV-cache entries, such as evicting KV-cache that will not be reused and protecting KV-cache that will be. Existing serving systems only schedule the matched prefix, losing potential for better performance.

From Deficiencies to Design Ignoring intra-prompt heterogeneity costs a serving system in two ways. First, shared tokens lose prefix caching: once a differing token appears ahead of them, their KV-cache can no longer be matched. Second, KV-cache protection and eviction lose precision: the system sees only a matched prefix and cannot manage entries by their individual reuse. Both costs are substantial. Among the agents we evaluated in §4, 51–92% of shared tokens miss the prefix cache because a differing token precedes them, and about 20% of served prompt tokens are used only once within a session. The layout freedom and the analyzability of intra-prompt heterogeneity address these deficiencies directly, upon which MORTIS is built.

3 DESIGN

In this section, we present the two-tier compiler-serving co-design of MORTIS. As shown in Figure 4, the *agent descriptor* is generated by combining analysis of the agent program with observation of runtime behaviors. The server side then exploits the derived descriptor to schedule the KV-cache efficiently, using *adaptive prompt re-layout* (§3.3) and *proactive KV-cache scheduling* (§3.4).

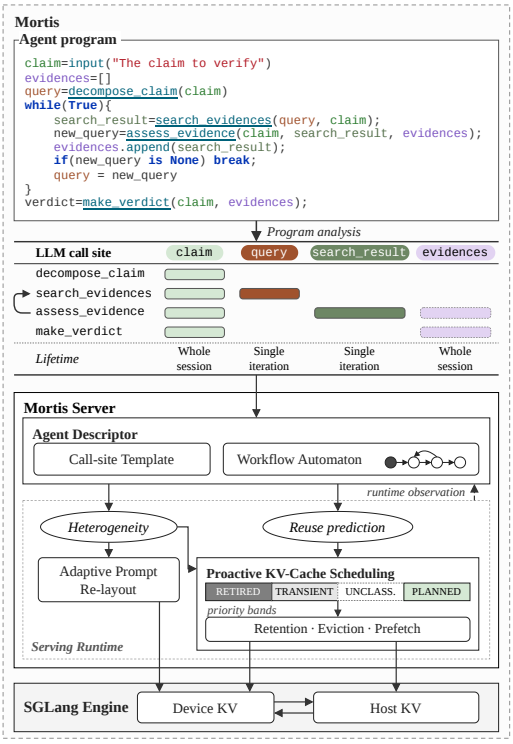


Figure 4. Architecture overview of MORTIS. By analyzing data that already resides in the program itself, MORTIS steers a more efficient KV-cache scheduling policy.

3.1 Agent Descriptor

The *agent descriptor* contains the metadata for scheduling. Static analysis provides its skeleton, and the server refines the weights with runtime observations. An agent descriptor contains two parts: (i) **Call-site Template:** It provides a heuristic for each component's lifetime and sharing pattern. (ii) **Workflow Automaton:** It contains the program-level structure of an agent, including workflow dependencies, execution time estimation and transition probabilities.

3.1.1 Call-site Template

Prompts can be decomposed into heterogeneous components. We classify these components into the following two categories: (i) **Invariant Specifications:** the components that are fixed once the program is compiled, such as the system prompt, response schema, and the tool declarations. They are rendered from the compiler's own intermediate representation and stay constant. (ii) **Dynamic Bindings:** because LLM calls are declared as ordinary functions, the arguments of a call-site are supplied at runtime. These runtime values are the dynamic bindings.

We define the call-site template as $\mathcal{T}_c = \langle \mathcal{I}_c,\ \mathcal{B}_c,\ \mathcal{H}_c \rangle$, where $\mathcal{I}_c$ denotes the invariant specifications, $\mathcal{B}_c$ the dynamic bindings, and $\mathcal{H}_c$ the heterogeneity annotation over those bindings.

Heterogeneity annotation For each dynamic binding, $\mathcal{H}_c$ records both how its value is shared and how long that value stays valid: $\mathcal{H}_c : \mathcal{B}_c \longrightarrow \mathcal{E} \times \mathcal{S}$, where $\mathcal{E}$ is the evolution component that governs which bytes a later prompt can reuse, and $\mathcal{S}$ is the scope of this binding in code, which decides when cached bytes can be given up.

We divide $\mathcal{E}$ into five classes, listed from the most to the least reusable: **(i) CONST:** the value is fixed by the program text and is identical at every call. **(ii) COPY:** the value is the same as the value another call-site carries. **(iii) RESP:** the value comes from the reply produced by another call-site. **(iv) EXTEND:** the binding is an append-only context, where the previous value remains a prefix of the new one. **(v) VOLATILE:** the value is freshly produced at each call, or its origin lies outside the analysis boundary.

3.1.2 Workflow Automaton

We abstract an agent's workflow as a time-annotated probabilistic finite-state automaton. We formulate it as:

$$\mathcal{A} = (Q, P, q_0, F, \mathcal{T}),$$

where: (i) Q is the set of states. Each state $q \in Q$ is labeled by an LLM call-site; (ii) P denotes the transition probabilities between states; (iii) $q_0 \in Q$ is the entrance call-site; (iv) F specifies the termination probability of the agent at each state; (v) $\mathcal{T} = \left\{ D_q^{\text{exec}}, D_{q,q'}^{\text{gap}} \right\}$ denotes the timing annotations associated with states and transitions, where D_q^{exec} is the execution-time distribution of the LLM call-site q, and $D_{q,q'}^{\text{gap}}$ is the inter-call gap distribution between the completion of call-site q and the invocation of its successor q'.

3.2 Recovering Heterogeneity via Compiler Analysis

The skeleton of the agent descriptor comes from program code: the graph structure of $\mathcal{A}$ and the tuple $\mathcal{T}_c$ for each call-site c.

Call-site graph construction The compiler identifies LLM call-sites from the typed AST. It then builds the interprocedural control-flow graph (CFG) of the agent and projects it onto these call-sites. An edge (q, q') is added whenever q' is directly reachable in one hop from q, while non-LLM function call-sites are contracted. If a call target cannot be resolved statically, the compiler conservatively adds edges to all candidate targets. This projected graph forms the unweighted structure of $\mathcal{A}$. Finally, the compiler applies Bourdoncle's weak topological ordering (Bourdoncle, 1993) to recover the loop nesting structure from the cycles in $\mathcal{A}$.

Separating $\mathcal{I}_c$ and $\mathcal{B}_c$ At runtime, the framework constructs a prompt by lowering the function declaration together with the runtime argument values. Because the declaration itself is fixed at compile time, the compiler runs the framework's lowering pass ahead of time using symbolic placeholders for the arguments. This produces the invariant specification $\mathcal{I}_c$, which is byte-identical to the static portion of the runtime prompt, while leaving one labeled placeholder per parameter to represent the dynamic bindings $\mathcal{B}_c$.

Recovering evolution classes ($\mathcal{E}$) For each binding $b \in \mathcal{B}_c$, the compiler computes the reaching definitions of its underlying variable across the CFG (including back edges, with local definitions shadowing earlier ones). Each reaching definition is classified by its right-hand side: (i) literals and module-level constants are classified as **CONST**; (ii) variables passed unmodified from an earlier call-site c' are classified as **COPY**(c'); (iii) values assigned from the return value of another call-site c' (traced via def–use chains) are classified as **RESP**(c'); (iv) prefix-preserving updates, like string/list concatenation ($v \leftarrow v + \Delta$), are classified as **EXTEND**; (v) all other definitions are defaulted to **VOLATILE**, such as external input.

If different CFG paths yield conflicting evolution classes for b, the compiler falls back to **VOLATILE**. To let the runtime resolve this conservativeness once actual transition (q, q') is known, the compiler attaches transition-specific metadata to each edge (q, q'). Specifically, it records which variables are modified along (q, q'), whether those modifications overwrite existing prefixes rather than appending to them, and the refined evolution class of each binding at q' along this specific edge (for example, refining an overall **VOLATILE** tag to **CONST** when a specific branch assigns a literal).

Recovering lifetime scopes ($\mathcal{S}$) For each binding b, the compiler records $\mathcal{S}(b)$ as the set of procedures (or loop bodies) that reassign or reset its underlying variable. If $\mathcal{S}(b) = \emptyset$, the variable is assigned only once and remains valid for the entire session. Otherwise, whenever execution enters a procedure in $\mathcal{S}(b)$, the variable is overwritten and its previously cached KV bytes become dead.

3.3 Adaptive Prompt Re-layout

MORTIS's *adaptive prompt re-layout* permutes prompt components to maximize prefix cache reuse.

3.3.1 Movable Components

First, the dynamic bindings $\mathcal{B}_c$ can be permuted freely. The compiler lowers each argument into a self-contained component labeled by its parameter name, and the framework places these components in declaration order only because that is the order of the argument list; re-layout may choose any order of the bindings.

Second, the instruction specification in $\mathcal{I}_c$ is movable. $\mathcal{I}_c$ consists of the system prompt, the response schema, the tool schema, and the instruction specification (such as task descriptions and parameter descriptions). The system prompt stays universally constant and should be placed at the very beginning, and the schemas are handled by the constrained decoding engine. The instruction specification can be put just after the system prompt or at the end of the prompt.

Algorithm 1 Layout decision algorithm

Require: Descriptor $(\mathcal{A}, \{\mathcal{T}_q\}_{q\in Q})$, switch *pin*, lengths ℓ
Ensure: Layouts $\{\mathcal{L}_q\}_{q\in Q}$
1: $B_{\mathrm{sh}} \leftarrow \{b : |\{q \in Q : b \in \mathcal{B}_q\}| > 1\}$ ▷ The set of all shared bindings
2: $G \leftarrow (B_{\mathrm{sh}}, \emptyset)$ ▷ precedence graph
3: **for all** $(q, q') \in \textsc{Edges}(\mathcal{A})$ **do**
4: $P_1, P_2, P_3 \leftarrow \emptyset$ Groups: unchanged, append, rewritten/one-sided
5: **for all** $b \in B_{\mathrm{sh}} \cap (\mathcal{B}_q \cup \mathcal{B}_{q'})$ **do**
6: $P_{\tau(b)} \leftarrow P_{\tau(b)} \cup \{b\}$ ▷ $\tau(b)$: group of b
7: $E(G) \leftarrow E(G) \cup \bigcup_{i<j} P_i \times P_j$
8: $\mathcal{C} \leftarrow \textsc{Tarjan}(G)$ ▷ SCCs of G, sets of bindings
9: $\hat{E} \leftarrow \{(C, C') \in \mathcal{C}^2 : C \neq C',\ (C \times C') \cap E(G) \neq \emptyset\}$
10: $\hat{G} \leftarrow (\mathcal{C}, \hat{E})$ ▷ condensation graph of G
11: $j(b) \leftarrow (-\ell(b), \mathit{depth}(b))$ for $b \in B_{\mathrm{sh}}$
12: $J(C) \leftarrow \min_{b\in C} j(b)$ for $C \in V(\hat{G})$
13: $L \leftarrow [\,]$
14: **while** $V(\hat{G}) \neq \emptyset$ **do**
15: $C^* \leftarrow \arg\min_{C\in V(\hat{G}),\ \mathit{indeg}(C)=0} J(C)$
16: $L \leftarrow L \,\|\, \textsc{Sort}(C^*, j)$; $\hat{G} \leftarrow \hat{G} - C^*$
17: **for all** $q \in Q$ **do**
18: $\pi_q \leftarrow [\, b \in L : b \in \mathcal{B}_q \,] \,\|\, \textsc{Sort}(\mathcal{B}_q \setminus B_{\mathrm{sh}}, K)$
19: $s_q \leftarrow \textsc{Spec}(\mathcal{I}_q)$ ▷ instruction specification of q
20: **if** $\neg \mathit{pin} \wedge \mathcal{B}_q \cap B_{\mathrm{sh}} \neq \emptyset$ **then**
21: $\mathcal{L}_q \leftarrow \pi_q \| s_q$
22: **else**
23: $\mathcal{L}_q \leftarrow s_q \| \pi_q$
24: **return** $\{\mathcal{L}_q\}_{q\in Q}$

3.3.2 Layout Decision

MORTIS decides the layout $\mathcal{L}_q$ for each call-site q by ranking the reusability of bindings (Algorithm 1). It places shared bindings ahead of private bindings and prioritizes stable bindings within both categories to maximize prefix cache efficiency.

Sorting shared bindings (lines 1–16) Shared bindings are those that are carried by multiple call-sites. For every

transition from q to q', we identify how each shared binding evolves. We categorize their behaviors into three sharing pattern groups, ordered as follows: **(i) unchanged:** Both call-sites carry it and its value does not change between the two calls. **(ii) append:** Both call-sites carry it and the program only appends to its value. **(iii) rewritten/one-sided:** The program rewrites it between the two calls, or only one call-site carries it.

As shown in Algorithm 1, MORTIS builds the precedence graph of all shared bindings based on the groups they belong to on each transition edge, condenses it into a DAG using Tarjan's algorithm, and finally performs a topological sort on the DAG. We key each SCC by the smallest **(*-length*, *depth*)** among its bindings, where: **(i)** *depth*: the number of transitions from the entrance to the first call-site that carries b; **(ii)** *length*: the token length of the value of b, observed by the server. The SCC with the smallest key goes first when multiple SCCs have zero in-degree. Each SCC is then expanded into the list of its bindings sorted also by **(*-length*, *depth*)**.

Sorting private bindings (line 18) The bindings carried by a single call-site are sorted in ascending lexicographic order of:

$$K(b) = \big(\textit{scoped},\ \rho,\ \textit{depth},\ -\textit{length}\big),$$

where: **(i)** *scoped*: the lifetime of b; 0 if it is session-wide ($\mathcal{S} = \emptyset$) and 1 otherwise; **(ii)** ρ: the stability rank for $\mathcal{E}$ in §3.1.1, from 1 for CONST to 5 for VOLATILE.

Placing the specification (lines 19–23) Placing the *instruction specification* at the end gives a longer common prefix. MORTIS places the specification at the end for each call-site that carries a shared binding. A switch keeps the specification at the front when the developer requires it.

3.4 Proactive KV-cache Scheduling

3.4.1 Reusable Head and Volatile Tail

A served prompt naturally divides into two segments: a reusable head and a volatile tail. The reusable head consists of two parts: a *static prefix* and a *session-shared extension*: (i) **Static Prefix** is the prefix that does not change, including $\mathcal{I}_c$ and the CONST bindings that re-layout has placed behind $\mathcal{I}_c$. (ii) **Session-shared Extension** consists of the bindings after the static prefix whose values recur unchanged in a later call of the same session.

3.4.2 Next-Call Prediction

MORTIS predicts three aspects of the next call: (i) The probability of the arrival. (ii) When the call will arrive. (iii) Which prefix of its prompt can be speculatively constructed.

Arrival probability Each call-site q can be located by a chain of loops enclosing it; for each call-site's invocation, we mark it with a tuple $\zeta = (i_1, i_2, \ldots, i_m)$, where $i_k, k \in [1, m]$ is the current iteration index of the k-th enclosing loop, from outermost to innermost.

MORTIS estimates the probability of the next call by a loop-tagged table. We define the table as $\mathcal{P}(q' \mid q, \zeta)$ and $\mathcal{F}(q, \zeta)$, where $\mathcal{P}$ is the probability that call-site q in context ζ is followed by call-site q'; $\mathcal{F}$ stands for the probability that the session ends at call-site q under context ζ. With an empty ζ (no loop context), $\mathcal{P}$ and $\mathcal{F}$ reduce to the base transition and termination probabilities P and F of the automaton $\mathcal{A}$.

The table is approximated and updated by frequency counts observed at runtime. Let q and ζ denote the call-site and loop context of the preceding call in the same session. When a call at q' arrives, the server increments the count of outcome q' in row (q, ζ); when the session closes at q, it increments the termination count in row (q, ζ). Each observation is also counted in every row (q, ζ_k) with $\zeta_k = (i_{m-k+1}, \ldots, i_m)$, $k = 0, \ldots, m$.

After a call at q lands, we compute the probability of each outcome o by Witten–Bell interpolation from the empty context ζ_0 to the full context $\zeta_m = \zeta$. The outcome o ranges over $\mathrm{succ}(q)$, the static successors of q plus a virtual outcome end for session termination:

$$\hat{P}_k(o) = \lambda_k \frac{c_k(o)}{n_k} + (1 - \lambda_k)\hat{P}_{k-1}(o), \quad k = 0, \ldots, m,$$

$$\hat{P}_{-1}(o) = \frac{1}{|\mathrm{succ}(q)|}, \qquad \lambda_k = \begin{cases} \dfrac{n_k}{n_k + d_k} & n_k > 0, \\ 0 & n_k = 0, \end{cases}$$

$c_k(o)$ is how many times outcome o followed a call at q in context ζ_k, $n_k = \sum_o c_k(o)$ is the total number of observations in that row, and d_k is the number of distinct outcomes with $c_k(o) > 0$. Then $\mathcal{P}(q' \mid q, \zeta) = \hat{P}_m(q')$ and $\mathcal{F}(q, \zeta) = \hat{P}_m(\mathrm{end})$. A context with no observation is skipped. The weight λ_k grows with n_k, so a frequently observed context dominates the prediction, while a rarely observed one defers to its shorter contexts.

Arrival time The time of arrival is conservatively approximated based on recorded statistics. The workflow automaton records $D^{\mathrm{gap}}_{q,q'}$ per transition: each time a call at q' arrives, the server appends the interval between the reply of the preceding call at q and this arrival. The interval is the agent's own time spent in its code between two calls.

We estimate the gap before the next call as:

$$\hat{g} = t_{50} - \tfrac{1}{2}(t_{90} - t_{50})$$

where t_{50} and t_{90} are the 50th and 90th percentiles of the gap samples on the transition (q, q'); the predicted call is then

expected to arrive $\hat{g}$ after the reply of the current call. The second term ensures that a transition with a more dispersed gap distribution receives a tighter estimate.

Prefix reconstruction MORTIS constructs the prefix of call-site q' for matching KV-cache entries to schedule. Since $\mathcal{H}_c$ records the relationships among arguments, MORTIS builds the prefix in layout $L_{q'}$ using the argument values observed in the current session. The construction stops at the first argument with an unknown value.

3.4.3 KV-cache Eviction Using Lifecycle Bands

Bands We categorize KV-cache entries into four bands, from first evicted to last: (1) **RETIRED:** the volatile tails of prompts belonging to completed sessions. (2) **TRANSIENT:** the volatile tail of a served prompt of a live session. (3) **UNCLASSIFIED:** entries that analysis cannot categorize. They keep the engine's default treatment. (4) **PLANNED:** prefixes that a predicted call will reuse.

Band transitions Bands are assigned at three moments of a session's lifecycle. (1) **When a served prompt enters the cache**, MORTIS splits it at its *reusable head* (§3.4.1). The volatile tail and the reply generated after it become *TRANSIENT*; the head stays *UNCLASSIFIED* until the next prediction refresh finds a successor call that starts with it, which tries to promote it to *PLANNED*. (2) **When a later call arrives**, the transition it takes may rewrite bindings that earlier heads carried. Each affected head is cut back to the prefix that survives the rewrite, and the bytes past it are demoted to *TRANSIENT*. (3) **When the session ends**, the volatile tails of every prompt it served drop to *RETIRED*. Their heads stay *UNCLASSIFIED* and age out by recency, since other sessions of the same program may still share them.

Band-ordered eviction Eviction empties the lowest band first and falls back to recency within the three lower bands. The radix tree's eviction heap is keyed by band, then by recency. The host tier keeps LRU eviction, except that *RETIRED* entries go first. Additionally, KV-cache entries marked as *TRANSIENT* are not admitted into the host tier as they will not be reused.

3.4.4 KV-cache Retention and Prefetch

A future reuse prediction is summarized as the following triple:

$$\textit{Prediction} = (\pi_{q'},\ \mathcal{P}(q' \mid q, \zeta),\ t_{\text{reply}} + \hat{g})$$

where $\pi_{q'}$ is the prefix reconstructed for q' and t_{reply} is the time the current call replied. Predictions are refreshed at every call, and are dropped when the predicted calls arrive or their predicted arrival time passes.

MORTIS acts on this triple in two ways: *Retention* keeps $\pi_{q'}$ from leaving the device, and *prefetch* restores the part of $\pi_{q'}$ that has already been demoted to the host.

Retention MORTIS ranks the predicted reusable KV-cache entries by the benefit of keeping $\pi_{q'}$ resident at every refresh:

$$\mathcal{D}(q') = \frac{\mathcal{P}(q' \mid q, \zeta) \cdot |\pi_{q'}|_{\text{cached}}}{\max(t_{\text{reply}} + \hat{g} - t_{\text{now}},\ \tau_{\min})}$$

where $|\pi_{q'}|_{\text{cached}}$ is the number of cached tokens of $\pi_{q'}$, and $\tau_{\min}$ is the floor that prevents an imminent call from receiving an unbounded $\mathcal{D}$. MORTIS admits predictions into *PLANNED* in descending order of $\mathcal{D}$ until they cover a fixed fraction of the device tier. Within *PLANNED*, each radix-cache node accumulates the $\mathcal{D}$ of every planned prefix that covers it, so a shared head outranks a private one.

Prefetch When part of $\pi_{q'}$ has been demoted to the host, the planner prefetches it back to the device ahead of the estimated arrival time. A prefetch release deadline is defined as:

$$t_{\text{release}} = t_{\text{reply}} + \hat{g} - \frac{|\pi_{q'}|_{\text{host}}}{B_{\text{h2d}}} - \frac{|\pi_{q'}|_{\text{uncached}}}{B_{\text{prefill}}}$$

where B_{h2d} and B_{prefill} are the measured host-to-device copy rate and prefill rate, $|\pi_{q'}|_{\text{host}}$ is the number of tokens of $\pi_{q'}$ cached only on the host, and $|\pi_{q'}|_{\text{uncached}}$ is the number of uncached tokens that require prefill.

Among released predictions, the earliest deadline goes first, with ties broken by the expected saving $\mathcal{P}(q' \mid q, \zeta) \cdot |\pi_{q'}|_{\text{cached}}$. A prefetch is released when the engine is idle or its deadline t_{release} approaches. It takes only free device memory, after reclaiming *RETIRED* and *TRANSIENT* entries. When no space is free, prefetching pauses but keeps refreshing the recency of these cache entries so that they do not age out.

4 EVALUATION

4.1 Experimental Setup

Testbed and model We implement MORTIS on top of SGLang v0.5.10.post1 and conduct evaluations on one A40 GPU using Qwen3-32B-AWQ. We allocate 15 GB of GPU memory to the device KV-cache pool and 25 GB of CPU memory to the host KV-cache pool.

Baselines We compare MORTIS against five baselines, reimplemented on our SGLang fork: **(i) SGLang** (Zheng et al., 2024), the vanilla SGLang engine with HiCache and default LRU eviction; **(ii) CacheScout** (Zhang et al., 2026), which learns call transitions online to prioritize and prefetch

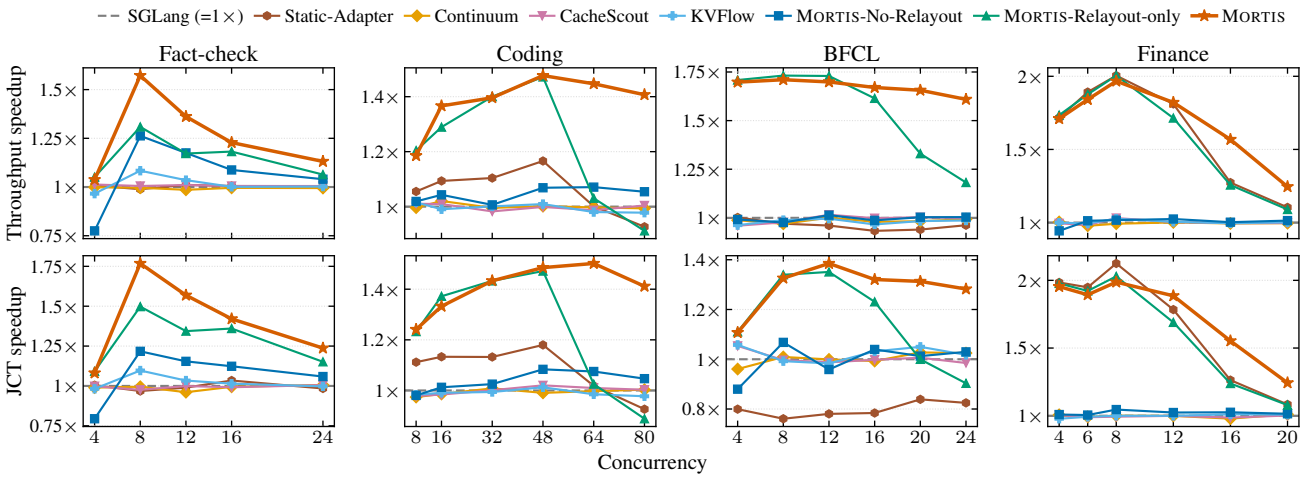


Figure 5. Throughput speedup (top) and average JCT speedup (bottom) over SGLang on each agent workload under varying concurrency.

agent prefixes; **(iii) Continuum** (Li et al., 2025), which pins agent KV-cache across tool gaps using adaptive TTLs; **(iv) KVFlow** (Pan et al., 2025), which schedules prefix eviction and prefetching via an Agent Step Graph; and **(v) Static-Adapter**, which mirrors DSPy's adapter (DSPy Contributors, 2026) by hoisting invariant components to the front under a fixed, program-agnostic layout.

We also evaluate the following ablation variants of MORTIS: (i) **MORTIS-Relayout-only** enables adaptive prompt re-layout but replaces proactive KV-cache scheduling with LRU eviction; (ii) **MORTIS-No-Relayout** enables proactive KV-cache scheduling but disables adaptive prompt re-layout; and (iii) **MORTIS-No-Prefetch** disables only KV-cache prefetching. Additionally, we run baseline serving systems on top of MORTIS's re-layout to isolate scheduling policy under the same prompt layout.

Workloads We evaluate MORTIS on four representative agent workloads, ported to Jac byLLM (Mars et al., 2023): a HoVer-style (Jiang et al., 2020) Fact-check agent, a HumanEval (Chen et al., 2021) coding agent, a BFCL (Patil et al., 2025) tool-use agent, and a FinanceBench (Islam et al., 2023) financial-analysis agent. We further evaluate a mixed workload comprising all four agent types to assess MORTIS's end-to-end performance under concurrent, heterogeneous demand.

4.2 End-to-End Performance of MORTIS

We test MORTIS under different numbers of concurrent workflows and on a mixed workload. Across these workloads, we compare the average job completion time (JCT) and throughput (prompt tokens served per minute).

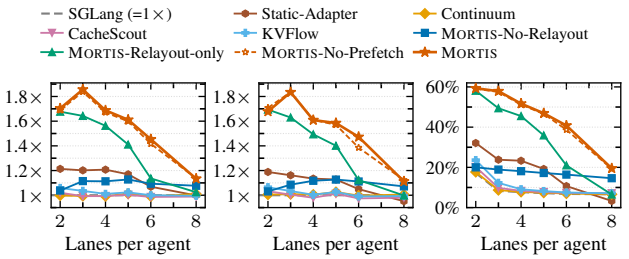
Performance on different workload types As shown in Figure 5, MORTIS achieves a 1.08–1.99× JCT speedup and increases throughput by 1.04–1.97× compared with SGLang, with geometric-mean gains of 1.45× in JCT and 1.49× in throughput. Across these agents, serving systems such as KVFlow show no significant improvement, because they ignore intra-prompt heterogeneity.

When concurrency is low, adaptive prompt re-layout accounts for nearly all of the improvement, because memory is sufficient. As load increases, however, the KV-cache becomes contested. Without knowing what to keep and what to give up, other serving systems fail to schedule it efficiently.

For the financial-analysis agent, MORTIS's improvement is small, because this agent is a simple pipeline that digests large blobs of documents. Under low pressure, adaptive prompt re-layout accounts for nearly all of the improvement by exposing larger prefix reuse. MORTIS's proactive KV-cache scheduling only takes effect under the highest load, because it can proactively evict stale KV-cache entries.

Performance on mixed workload We evaluate MORTIS under varying pressure on the mixed workload and report metrics over each system's steady-state window. We vary the number of lanes (concurrent workflows) per agent, keeping it equal across all four agents.

As shown in Figure 6, MORTIS achieves up to 1.86× throughput speedup and 1.83× JCT speedup over SGLang at 3 lanes per agent. Under low pressure, adaptive prompt re-layout alone accounts for nearly all of the gain, since device memory is still large enough to retain the reusable KV-cache. As pressure grows, the cache becomes contested and the gain from re-layout alone shrinks, while proactive KV-cache scheduling retains a clear advantage by protecting the reusable heads that LRU would evict. Under the heaviest



(a) Throughput speedup (b) JCT speedup (c) Cache hit rate

Figure 6. Throughput speedup and JCT speedup over SGLang, and cache hit rate, under varying lanes per agent on the mixed workload. JCT speedup is the geometric mean of the four agents' speedups.

Table 1. Task accuracy under three prompt layouts that appear in our evaluation.

Prompt layout	Fact-check	Coding	Financial-analysis	BFCL
Original byLLM	16.3%	55.3%	34.9%	14.1%
MORTIS	21.9%	54.8%	37.2%	18.7%
Static-Adapter	16.3%	52.2%	37.8%	16.0%

pressure, configurations converge toward the baseline, as the engine is fully saturated and no free memory is left for cache management to exploit. Across the six lane counts, MORTIS achieves geometric-mean speedups of 1.56× in throughput and 1.53× in JCT.

In the mixed workload, agents have heterogeneous demands for KV-cache memory, which reduces the chance that they contend for memory at the same time. By exploiting intra-prompt heterogeneity, MORTIS can promptly yield stale KV-cache to other requests that need memory, which further amplifies the benefit of MORTIS.

Task quality The performance gains of MORTIS do not come at the cost of task quality. As shown in Table 1, across three different layouts produced by the original byLLM, Static-Adapter, and MORTIS, reordering components does not harm task quality.

4.3 Cache Efficiency of MORTIS

Figure 7 shows the cache hit rate of prompt tokens at the same concurrency levels as Figure 5. On Coding and BFCL, adaptive prompt re-layout alone brings the hit rate at low concurrency to that of MORTIS: it places the stable and shared bindings at the front, where they form a reusable head that later calls hit. As concurrency grows, the hit rate of MORTIS-Relayout-only drops quickly, because LRU eviction does not distinguish reusable heads from volatile tails. With proactive KV-cache scheduling, MORTIS keeps a higher hit rate than MORTIS-Relayout-only as concurrency grows, and on Coding its hit rate stays nearly flat. On the

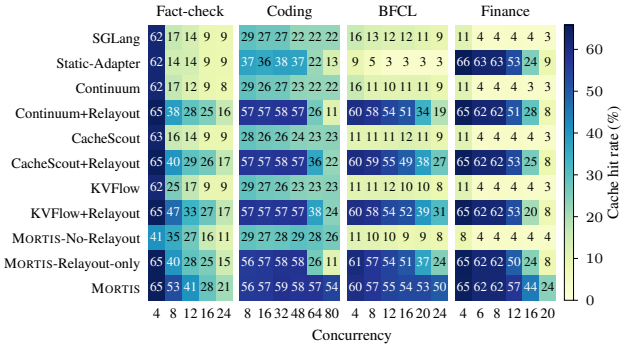


Figure 7. Cache hit rate (%) of prompt tokens under varying numbers of concurrent workflows.

Fact-check agent, MORTIS's hit rate exceeds that of MORTIS-Relayout-only from the lowest concurrency. Continuum and CacheScout stay close to SGLang at every concurrency. MORTIS-No-Relayout stays below MORTIS, since under the original prompt order the reusable head is short. The mixed workload shows a similar pattern (Figure 6).

4.4 Ablation Study

As the geometric-mean speedups of JCT and throughput over SGLang on the four workloads in Table 2 show, *adaptive prompt re-layout* and *proactive KV-cache scheduling* are superadditive: when both are enabled, the speedup is higher than the product of the two individual speedups. The same effect appears on the mixed workload (Figure 6a), where adding proactive KV-cache scheduling on top of adaptive prompt re-layout substantially improves performance.

The gain of MORTIS's KV-cache scheduling builds on re-layout, because re-laying out the prompt order exposes a reusable prefix and a suffix that can be abandoned. MORTIS's KV-cache scheduling substantially improves performance on top of re-layout, but scheduling alone (MORTIS-No-Relayout) barely helps (Table 2). The same trend appears in Figures 6 and 5.

Prefetching contributes little on this testbed. As shown in Figure 6a, disabling it (MORTIS-No-Prefetch) lowers throughput by at most 2.6% across the six lane counts. The gain of proactive KV-cache scheduling mainly comes from retention, which keeps the reusable heads resident.

4.5 Comparison under Identical Layouts

Serving the baseline systems with re-laid prompts identical to MORTIS's, we evaluate their speedup against MORTIS-Relayout-only. As shown in Figure 8a, these systems show marginal improvement over MORTIS-Relayout-only, but are still beaten by MORTIS. We further evaluate on the four agent workloads separately. Figure 8b shows the same trend

Table 2. Geometric-mean speedup over SGLang across all concurrency levels of the four agent workloads, with MORTIS's *Adaptive Prompt Re-layout* and *Proactive KV-cache Scheduling* enabled or disabled.

(a) JCT speedup

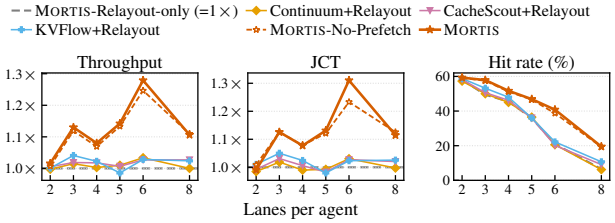
	Scheduling	
Re-layout	Off	On
Off	1.00×	1.03×
On	1.30×	1.45×

(b) Throughput speedup

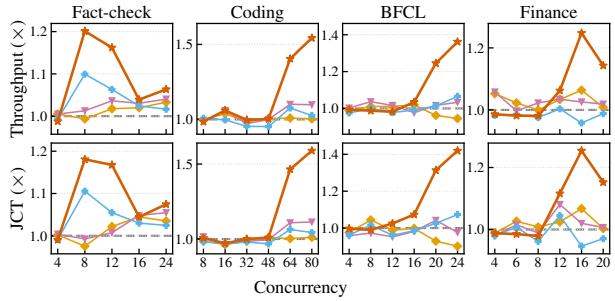
	Scheduling	
Re-layout	Off	On
Off	1.00×	1.02×
On	1.36×	1.49×

as in Figure 8a. The largest gains are from CacheScout on Coding and KVFlow on Fact-check, both up to 1.11× in JCT, but MORTIS still outperforms all of them.

This deficiency in baseline serving systems arises because they manage the KV-cache only at the granularity of a prefix and its suffix, which ignores intra-prompt heterogeneity. They cannot identify precisely which KV-cache entries will be reused, leading to inefficient KV-cache scheduling.



(a) Throughput, JCT speedup and hit rate on the mixed workload



(b) Throughput and JCT speedup on each agent workload

Figure 8. Baseline systems served with MORTIS's re-laid prompts. Speedups are measured against MORTIS-Relayout-only.

4.6 Sensitivity Study

We further run the mixed workload on four A40 GPUs with TP=4. Keeping the same KV-cache pool size, TP=4 increases the prefill speed by 1.74× (from 1292.4 to 2251.8 tokens/s), which lowers the cost of a cache miss.

Even when cache misses become cheaper under TP=4, MORTIS maintains its advantage (Figure 9), achieving geometric-mean speedups of 1.60× in throughput and 1.58× in JCT (up to 1.92× and 1.87×).

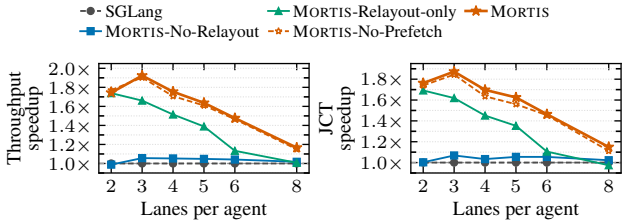


Figure 9. Speedup over SGLang under the mixed workload with A40 TP=4.

5 Related Work

Application-level information for serving Parrot (Lin et al., 2024) represents intermediate results as semantic variables and uses their producer–consumer relationships for scheduling and prefix-aware placement. Teola (Tan et al., 2025) exposes operators finer than a call, removes unnecessary dependencies, computes available prefixes while upstream operators produce the remaining inputs, and forwards partial outputs downstream.

Prompt construction for prefix reuse DSPy's adapters and NOOA fix a component order by convention or by developer labels (DSPy Contributors, 2026; Furgale et al., 2026), and the SGLang frontend moves constant components forward at compile time (Zheng et al., 2024). For relational analytics, Liu et al. (Liu et al., 2025) order rows and the fields within them jointly, grouping rows that share field values and using functional dependencies to expose further sharing. Reuse can also go beyond matching prefixes: Prompt Cache (Gim et al., 2024) precomputes prompt modules at prescribed positions, and CacheBlend (Yao et al., 2025) recomputes the tokens whose attention crosses chunk boundaries when combining cached chunks.

KV-cache scheduling across execution gaps Serving systems also steer the KV-cache according to tool-call gaps in ReAct agents. InferCept (Abhyankar et al., 2024) weighs the cost of keeping, swapping, and recomputing KV-cache per interruption to make scheduling decisions; Continuum (Li et al., 2025) holds the idle KV-cache for a time-to-live derived from observed inter-invocation delays; TokenCake (Bian et al., 2026) estimates the tool call's duration, offloads idle KV to the host, and restores it before the expected return.

6 Conclusion

MORTIS exploits intra-prompt heterogeneity via runtime-assisted static analysis, steering KV-cache management through adaptive prompt re-layout and proactive scheduling. Evaluations show MORTIS achieves up to 1.99× JCT speedup and 1.97× higher throughput over SGLang.

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